MyWalk

Habit Categories — Implementation Plan

Version 1.0 · Flutter / Firebase · Claude Code Brief

1. Overview & Objectives

This document specifies the full implementation of the new Habit Category system for MyWalk. It is a Claude Code brief — every section is written to be directly actionable by the developer without further clarification.

> **What is changing**
> The current model (one Category = one Habit) is being replaced with a two-level hierarchy: Categories contain Sub-categories, which in turn contain individual Habits. The user experience on the Today screen and the habit creation flow both change, but the underlying Habit data model is extended rather than replaced.

1.1 Mental Model Change

Before

Exercise → is the Habit
Scripture → is the Habit
Fasting → is the Habit

After

Spiritual Disciplines → Category
  Prayer → Sub-category
  My daily morning prayer → Habit

1.2 Principles

- Zero redesign of the Set It Up form — it is unchanged
- Zero redesign of the Today screen interaction model
- Categories are a discovery/creation tool only — not a runtime container
- The Today screen remains flat: individual habit cards, grouped by category header
- A sub-category picker screen is inserted between the category grid and the Set It Up form
- Existing habits are migrated to the new model without data loss

2. The 12 Categories

Categories are presented in the "Loving God → Loving Self → Loving Others → Your Walk" framework, with section dividers in the UI. The ordering and grouping is fixed and not user-reorderable.

Group A — Loving God

#	Category	Key Sub-categories
1	**Spiritual Disciplines**	Prayer, Scripture Reading, Scripture Memorisation, Worship, Fasting, Church Attendance, Communion, Confession, Intercession, Evangelism, Pilgrimage/Retreat
2	**Contemplation & Reflection**	Journaling, Gratitude Journaling, Silence, Solitude, Meditation, Lectio Divina, Examen, Breath Prayer, Spiritual Direction, Life Review, Therapy/Counselling
3	**Exercise**	Strength Training, Cardio, Walking, Flexibility & Mobility, Sports, Martial Arts, Dance
4	**Health & Nutrition**	Hydration, Healthy Eating, No Sugar, No Alcohol, Vitamins & Supplements, Medication, No Smoking, Dental Hygiene, Doctor Visits
5	**Rest & Renewal**	Sleep, Sabbath, Digital Sabbath, No Social Media, Time in Nature, Leisure, Unplugging
6	**Reading, Learning & Study**	Bible Study, Theology, Christian Living, General Non-fiction, Fiction, Podcasts, Online Courses, Language Learning, Professional Development
7	**Connection**	Quality Time with Family, Marriage Investment, Parenting, Friendship, Mentorship, Discipleship, Community/Church, Accountability, Hospitality
8	**Service & Generosity**	Volunteering, Acts of Kindness, Tithing, Charitable Giving, Serving at Church, Advocacy, Meals for Others, Visiting the Lonely
9	**Stewardship**	Budgeting, Saving, Debt Reduction, No Impulse Buying, Time Planning, Decluttering, Environmental Care, Home Maintenance, Digital Organisation
10	**Creativity**	Visual Art, Photography, Writing, Music, Crafts, Cooking/Baking, Gardening, Design, Film/Video, Dance
11	**Habit Breaking**	No Alcohol, No Pornography, No Smoking, No Gambling, No Social Media, No Gossip, No Negative Self-talk, No Overworking, No Complaining, Custom
12	**Create My Own**	Fully custom — user defines name, description, tracking type, schedule, icon, colour, Fruit tags

3. Data Model Changes

3.1 New Firestore Collections

habit_categories (static — seeded at app init)

A top-level collection containing the 12 fixed categories. This is read-only from the client. Seeded via a Cloud Function on first deploy.

Field	Type	Description
id	`String`	Snake case ID e.g. spiritual_disciplines, exercise
name	`String`	Display name e.g. "Spiritual Disciplines"
description	`String`	One-sentence description shown on category card
icon	`String`	Icon asset name from app icon set
colour	`String`	Hex colour for category card background
group	`String`	One of: loving_god, loving_self, loving_others, your_walk
groupLabel	`String`	Display label for the group divider
sortOrder	`int`	Position within the grid (1–12)
isCustom	`bool`	True only for the "Create My Own" category

habit_subcategories (static — seeded at app init)

A top-level collection containing all sub-categories. Each document references its parent category.

Field	Type	Description
id	`String`	Snake case ID e.g. prayer, scripture_reading
categoryId	`String`	Reference to parent habit_categories document
name	`String`	Display name e.g. "Prayer"
description	`String`	Brief description shown on sub-category picker card
icon	`String`	Icon asset name (inherits from category if not set)
trackingTypeSuggestion	`String`	Suggested default: check_in, timed, count, abstain
defaultTargetMinutes	`int?`	Suggested default target for timed habits (nullable)
sortOrder	`int`	Position within the sub-category picker grid
isCustom	`bool`	True only for "+ Custom" option within each category

users/{uid}/habits/{habitId} (extended — not replaced)

The existing habit document is extended with two new fields. All existing fields remain unchanged.

Field	Type	Description
categoryId	`String`	NEW — references habit_categories. Required on all new habits.
subcategoryId	`String`	NEW — references habit_subcategories. Required on all new habits.
categoryName	`String`	NEW — denormalised display name for grouping on Today screen

subcategoryName	`String`	NEW — denormalised display name for habit card subtitle

Migration Note

Existing habits without categoryId/subcategoryId should be assigned category "exercise", "scripture_study", etc. during a one-time migration Cloud Function. Run this on app launch if the user has habits without a categoryId. See Section 7 for migration details.

4. Navigation Flow

4.1 New Route — Category Grid Screen

This screen already exists. No changes required to the grid itself. The only change is what happens when a tile is tapped.

> **Before**
> Category tile tapped → Set It Up screen (with category pre-filled)

> **After**
> Category tile tapped → Sub-category Picker screen → Set It Up screen (with category AND sub-category pre-filled)

> *ℹ Exception: "Create My Own" category bypasses the sub-category picker and goes directly to Set It Up with both fields empty and editable.*

4.2 New Screen — Sub-category Picker

Route: /habits/new/subcategory

This is a new screen inserted between the Category Grid and the Set It Up form.

Layout

- App bar: back arrow + category name as title + "Cancel" action
- Subheading: "What would you like to practise?" (existing app copy pattern)
- Grid of sub-category cards — 2 columns, same visual style as category grid
- Each card: icon (top) + name (below) + brief description (small, below name)
- Final card in every category: "+ Custom" — opens Set It Up with subcategoryId = custom and name field focused and empty
- No search bar needed — maximum sub-categories in any category is 12

Behaviour

- Tapping a sub-category card navigates to Set It Up with categoryId and subcategoryId passed as route arguments
- Both values are pre-filled and shown as read-only chips at the top of the Set It Up form
- User can tap the chips to go back and change their selection
- Back arrow returns to Category Grid

4.3 Updated Screen — Set It Up

The Set It Up form is functionally unchanged. The following additions are made:

- NEW: Read-only "Category" chip displayed at top of form — shows categoryName
- NEW: Read-only "Sub-category" chip displayed at top of form — shows subcategoryName
- Both chips have a small edit icon (pencil) that navigates back to sub-category picker
- categoryId and subcategoryId are saved to the habit document on submission
- All other fields unchanged

> *ℹ The Habit Name field should be pre-populated with the sub-category name as a starting suggestion, which the user can edit. E.g. selecting "Prayer" pre-fills the name with "Prayer". This reduces friction.*

5. Today Screen Changes

5.1 Grouped Habit Cards

The Today screen (Give tab) displays habit cards in a flat list, grouped by category with a subtle section header above each group. The interaction model of each card is unchanged.

Sorting Logic

1. Daily Gratitude always first (existing behaviour — unchanged)
2. Remaining habits grouped by categoryId, sorted by the category's sortOrder field
3. Within each category group, habits sorted by their own createdAt timestamp (oldest first)

Category Group Header

Rendered above the first habit card of each category group. Not rendered above Daily Gratitude.

- A thin horizontal rule spanning the full width
- Category name in small caps or uppercase tracking — muted, not prominent
- Use the app's existing text style: ~11px, letter-spacing 1.5, color: rgba(255,255,255,0.4)
- No icon in the header — keep it minimal
- Spacing: 16px above the rule, 8px below the rule, then the first habit card

Habit Card — Subtitle Addition

Each habit card gains a subtitle line showing the sub-category name, rendered below the habit name in a smaller, muted style. This helps the user distinguish e.g. "Morning prayer" from "Evening prayer" — both under Prayer — at a glance.

- Subtitle: subcategoryName — e.g. "Prayer", "Scripture Reading"
- Style: 11px, muted gold or rgba(255,255,255,0.45), no icon
- Only shown if the habit has a subcategoryId set (existing habits without one show nothing)

5.2 No Change to Card Interaction

Tapping a habit card opens the existing habit detail/log screen. This is unchanged. The category grouping is visual only — it does not alter any tap behaviour, swipe behaviour, or check-in logic.

6. State Management

6.1 New Repositories

HabitCategoryRepository

Reads from habit_categories collection. Cached locally — this data never changes at runtime.

- getCategories() → List<HabitCategory> — returns all 12 sorted by sortOrder
- getCategoryById(String id) → HabitCategory?
- getCategoriesByGroup(String group) → List<HabitCategory>

HabitSubcategoryRepository

Reads from habit_subcategories collection. Cached locally.

- getSubcategories(String categoryId) → List<HabitSubcategory> — returns all for a category sorted by sortOrder
- getSubcategoryById(String id) → HabitSubcategory?

6.2 New Domain Entities

HabitCategory

```
class HabitCategory {
  final String id;
  final String name;
  final String description;
  final String icon;
  final String colour;
  final String group;
  final String groupLabel;
  final int sortOrder;
  final bool isCustom;
}
```

HabitSubcategory

```
class HabitSubcategory {
  final String id;
  final String categoryId;
  final String name;
  final String description;
  final String? icon;
  final String trackingTypeSuggestion;
  final int? defaultTargetMinutes;
  final int sortOrder;
  final bool isCustom;
}
```

Habit (existing — extended)

Add to the existing Habit entity and its fromFirestore/toFirestore methods:

```
final String? categoryId;       // NEW
final String? subcategoryId;    // NEW
```

```
final String? categoryName;     // NEW — denormalised
final String? subcategoryName;  // NEW — denormalised
```

6.3 New Cubits / Providers

HabitCategoryCubit

- State: HabitCategoryState (loading / loaded / error)
- Loads all categories and sub-categories on app init
- Exposed at root MultiProvider so it is available app-wide

SubcategoryPickerCubit

- Scoped to the sub-category picker screen
- Receives categoryId on init
- Loads sub-categories for that category
- Emits selected subcategoryId on tap

7. Migration of Existing Habits

7.1 Strategy

Existing habits were created before the category system existed. They have a category field (String) matching the old 10-category names. The migration maps these to the new categoryId and subcategoryId fields.

7.2 Mapping Table

Old category	New categoryId	New subcategoryId
`exercise`	`exercise`	custom (user to specify)
`scripture`	`spiritual_disciplines`	scripture_reading
`rest`	`rest_and_renewal`	sleep
`fasting`	`spiritual_disciplines`	fasting
`study`	`reading_learning_study`	custom
`service`	`service_and_generosity`	custom
`connection`	`connection`	custom
`health`	`health_and_nutrition`	custom
`abstain`	`habit_breaking`	custom
`gratitude (built-in)`	`contemplation_reflection`	gratitude_journaling

> **subcategoryId = "custom"**
> Where subcategoryId is mapped to "custom", the habit name already captures what the habit is. The user can optionally update it through the habit edit screen. Do not prompt them during migration — it is disruptive.

7.3 Migration Cloud Function

Create a callable Cloud Function: migrateHabitCategories

- Called once on app launch if any habit document is missing categoryId
- Reads all habits for the user
- Applies the mapping table above
- Writes categoryId, subcategoryId, categoryName, subcategoryName to each habit document in a batched write
- Marks migration complete in users/{uid}/state/preferences as habitCategoryMigrationComplete: true
- Function is idempotent — safe to call multiple times
- Run migration before loading the Today screen on first launch after update

8. Seeding the Static Data

8.1 Cloud Function: seedHabitCategories

A one-time Cloud Function that populates habit_categories and habit_subcategories. Run this on first deploy. Guard with an existence check so it is safe to run again.

- Check if habit_categories collection has documents — if yes, skip
- Write all 12 category documents
- Write all sub-category documents in batched writes (500 per batch max)
- Log completion

8.2 Local Fallback

Bundle a local JSON seed file (assets/data/habit_categories.json) as a fallback for offline first launch. The HabitCategoryRepository checks Firestore first; if unavailable, loads from the asset bundle.

9. Screen Specifications

9.1 Category Grid Screen (existing — minor update)

File: lib/presentation/habits/category_grid_screen.dart

- CHANGE: On category tile tap, navigate to SubcategoryPickerScreen passing categoryId
- EXCEPTION: If categoryId == "create_my_own", navigate directly to HabitSetupScreen with empty args
- CHANGE: Add group section dividers between groups using the groupLabel field
- Group divider style: small uppercase label, thin gold rule, 24px vertical spacing
- NO OTHER CHANGES

9.2 Sub-category Picker Screen (new)

File: lib/presentation/habits/subcategory_picker_screen.dart

- Route: /habits/new/subcategory
- Constructor args: String categoryId
- Uses SubcategoryPickerCubit scoped to this screen
- App bar: BackButton + Text(category.name) + TextButton("Cancel") that pops to root

Body layout:

- 16px horizontal padding
- Subheading: "What would you like to practise?" — same style as category grid subheading
- 12px gap
- GridView.builder — 2 columns, childAspectRatio: 1.1, crossAxisSpacing: 12, mainAxisSpacing: 12

Sub-category card:

- Background: same charcoal card colour as category grid tiles
- Icon: 28px, gold, top-centre
- Name: 14px, white, medium weight, below icon
- Description: 11px, rgba(255,255,255,0.5), italic, below name, max 2 lines
- Tap: call cubit.selectSubcategory(id) then navigate to HabitSetupScreen

Custom card (always last):

- Icon: plus circle
- Name: "Custom"
- Description: "Name it yourself"
- Tap: navigate to HabitSetupScreen with subcategoryId = "custom", name field empty and focused

9.3 Habit Setup Screen (existing — minor update)

File: lib/presentation/habits/habit_setup_screen.dart

Constructor args (add):

- String categoryId
- String subcategoryId
- String categoryName
- String subcategoryName

UI additions (insert above the Habit Name field):

- Row of two read-only chip widgets: [categoryName chip] [subcategoryName chip]
- Each chip has a small pencil icon on the right
- Tapping either chip navigates back to SubcategoryPickerScreen (not all the way to Category Grid)
- Chip style: dark background, gold border 0.5px, gold text, 12px font, 8px horizontal padding

Pre-population:

- Habit Name field pre-populated with subcategoryName if subcategoryId != "custom"
- If subcategoryId == "custom", Habit Name field is empty and focused

On save:

- Write categoryId, subcategoryId, categoryName, subcategoryName to the habit document
- All existing save logic unchanged

9.4 Today Screen (existing — minor update)

File: lib/presentation/give/give_screen.dart (or equivalent)

Sorting:

- Sort habits by: (1) isGratitude desc, (2) category.sortOrder asc, (3) habit.createdAt asc
- Build a grouped list: Map<String, List<Habit>> keyed on categoryId

Rendering:

- For each category group, render a CategoryGroupHeader widget above the first card
- CategoryGroupHeader: thin horizontal Divider (gold, 0.5px) + Text(categoryName) in small caps style
- Do not render CategoryGroupHeader above the Gratitude card

Habit card subtitle (minor change to existing card widget):

- Add subtitle text below habit name: subcategoryName
- Style: 11px, rgba(255,255,255,0.4), no icon
- Only render if habit.subcategoryName != null

10. File Structure

New files to create:

```
lib/data/repositories/habit_category_repository.dart
lib/data/repositories/habit_subcategory_repository.dart
lib/domain/entities/habit_category.dart
lib/domain/entities/habit_subcategory.dart
lib/domain/repositories/i_habit_category_repository.dart
lib/domain/repositories/i_habit_subcategory_repository.dart
lib/presentation/habits/subcategory_picker_screen.dart
lib/presentation/habits/cubit/subcategory_picker_cubit.dart
lib/presentation/habits/cubit/subcategory_picker_state.dart
lib/presentation/habits/widgets/subcategory_card.dart
lib/presentation/give/widgets/category_group_header.dart
assets/data/habit_categories.json
functions/src/seedHabitCategories.ts
functions/src/migrateHabitCategories.ts
```

Files to modify:

```
lib/presentation/habits/category_grid_screen.dart   (navigation change + group
dividers)
lib/presentation/habits/habit_setup_screen.dart      (category chips + pre-population)
lib/presentation/give/give_screen.dart               (grouping + headers + card
subtitle)
lib/domain/entities/habit.dart                       (add 4 new fields)
lib/data/repositories/habit_repository.dart          (serialise new fields)
lib/main.dart                                         (register new providers)
```

11. Implementation Order

Implement in this sequence to avoid breaking existing functionality:

Phase 1 — Data Foundation (no UI changes)

4. Create HabitCategory and HabitSubcategory domain entities
5. Create repository interfaces and implementations
6. Create assets/data/habit_categories.json with all 12 categories and all sub-categories
7. Implement HabitCategoryCubit and register at root MultiProvider
8. Extend Habit entity with 4 new nullable fields
9. Update habit_repository fromFirestore/toFirestore for new fields
10. Write and deploy seedHabitCategories Cloud Function
11. Write and deploy migrateHabitCategories Cloud Function
12. Test: verify seed data reads correctly, verify migration runs without errors

Phase 2 — New Sub-category Picker Screen

13. Create SubcategoryPickerCubit and state
14. Create SubcategoryCardWidget
15. Create SubcategoryPickerScreen with full layout
16. Register route in GoRouter
17. Update CategoryGridScreen to navigate to SubcategoryPickerScreen on tile tap
18. Test: full flow from category grid → sub-category picker → Set It Up

Phase 3 — Set It Up Screen Updates

19. Add category/subcategory chip display to HabitSetupScreen
20. Add pre-population of Habit Name field
21. Wire up chip tap → back to SubcategoryPickerScreen
22. Update save logic to write categoryId, subcategoryId, categoryName, subcategoryName
23. Test: create a new habit end-to-end, verify all fields saved to Firestore

Phase 4 — Today Screen Updates

24. Create CategoryGroupHeader widget
25. Update GiveScreen to sort and group habits by category
26. Add group headers between category groups
27. Add subcategoryName subtitle to habit cards
28. Test: Today screen renders correctly with mixed old and new habits

Phase 5 — Migration & Polish

29. Call migrateHabitCategories on app launch if migration not yet complete
30. Verify all existing habits display correctly on Today screen after migration
31. Handle edge cases: user with no habits, user with only Gratitude, offline user
32. Regression test: check-in flow, week score, fruit portfolio, Journey screen all still correct
33. Remove any dead code from old single-level category system

12. Edge Cases & Constraints

Scenario	Type	Required Behaviour
User has no habits	`Edge`	Today screen shows only Gratitude card — no category headers
User has only habits in one category	`Edge`	One category header shown — works fine
User offline on first launch after update	`Constraint`	Load categories from assets/data/habit_categories.json fallback. Queue migration for when connection returns.
Migration fails mid-way	`Error`	Migration is idempotent — re-run on next launch. Do not block UI.
Sub-category "custom" selected	`Edge`	Habit Name field empty, focused. subcategoryName saved as the user's custom habit name after they type it.
Habit edited after migration	`Edge`	Edit screen shows existing category chips. User can change sub-category if desired but is not forced to.
Fruit of the Spirit tagging	`Constraint`	Unchanged. Fruit tags are applied per individual habit, not per category.
SOS button	`Constraint`	Unchanged. SOS reads the individual habit's purpose and anchor verse.
Prayer Circles shared habits	`Constraint`	Circle habits also gain categoryId and subcategoryId fields. Seed with spiritual_disciplines / prayer as default.
Free plan (2 habit limit)	`Constraint`	Limit applies to total habits regardless of category. Logic unchanged.
Week cycle & dedication	`Constraint`	Unchanged. Dedication applies to all habits. Category grouping is display-only.

13. Acceptance Criteria

The implementation is complete when all of the following pass:

Category Grid

- Grid displays 12 categories in the correct order with group dividers
- Tapping any category except "Create My Own" opens Sub-category Picker
- Tapping "Create My Own" opens Set It Up directly with empty fields

Sub-category Picker

- Displays correct sub-categories for each of the 12 categories
- Final card in each category is always "+ Custom"
- Tapping a sub-category card opens Set It Up with correct pre-filled values
- Tapping "+ Custom" opens Set It Up with empty name field, focused
- Cancel returns to Today screen, not just to Category Grid

Set It Up

- Category and Sub-category chips display correctly
- Habit Name pre-populated with sub-category name (except custom)
- Tapping chips returns to Sub-category Picker (not Category Grid)
- Saved habit document in Firestore contains all 4 new fields
- Existing fields and save behaviour unchanged

Today Screen

- Habits grouped by category with correct section headers
- Gratitude card always first with no section header above it
- Category headers use correct category names
- Habit cards show subcategoryName as subtitle
- Habits without subcategoryName (pre-migration) show no subtitle — no error
- Check-in, retroactive logging, SOS, week score all work as before

Migration

- All existing habits have categoryId and subcategoryId after migration
- Migration runs only once (guarded by habitCategoryMigrationComplete flag)
- Migration is safe to run multiple times without duplicating data
- User with zero habits — migration completes without error